UNIT 8

ELECTRIC FIELD

ELECTROSTATIC FORCE:

"The attractive or repulsive force between charges is called electrostatic or coulomb's force."

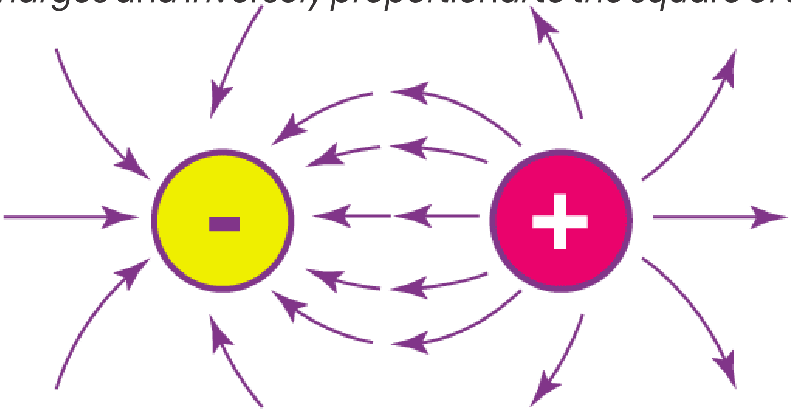
COULOMB'S LAW:

Introduction:

In 1785, Charles Augustine de Coulomb, French physicist, investigated quantitatively the force between localized (Point) charges. He used a "Torsion balance" to arrive at the inverse square law for an electrostatic force.

Statement:

"Same charges repel each other and different charges attract each other with a force which is known as electrostatic force, this force is directly proportional to the product of magnitude of charges and inversely proportional to the square of the distance between them."



Mathematically:

Consider two point charges q1q_1q1 and q2q_2q2 placed at distance rrr from each other, Mathematically the electrostatic force between them is,

$$F \propto q_1\ q_2$$

$$F \propto \frac{1}{r^2}$$

By combining equation (i) and (ii),

$$F \propto \frac{q^1\ q^2}{r^2}$$

$$F = k\,\frac{q^1 q^2}{r^2}$$

Where 'k' is the constant of proportionality and it depends on:

• The nature of the medium between the charges.

• The system of units chosen.

Vector form of Coulomb's law: If F is the unit vector along the line joining two charges from q1 to q2, then,

Coulomb's force between the charges in the presence of free space: If the medium is free space or vacuum and if M.K.S system is chosen, then the value of k is:

$$k = \frac{1}{4\pi\epsilon_0}$$

Where ϵ0 (Epsilon nought) is called the permittivity of free space; its value is:

$$\epsilon_0 = 8.85\times10^{-12} C2N^{-1}m^{-2}$$

The value of k becomes:

$$k = \frac{1}{4\pi\epsilon_0} = 9\times10^{9} Nm^2C^{-2}$$

Hence, Coulomb's law is:

$$k = \frac{1}{4\pi\epsilon_0}\frac{q^1\, q^2}{r^2}$$

Coulomb's force between the charges in the presence of medium (Dielectric): If the medium is dielectric and M.K.S system is chosen, then the value of k is:

$$k = \frac{1}{4\pi\epsilon_0\epsilon_r}$$

Where ϵr (Epsilon r) is called relative permittivity and its value depends upon the nature of the medium placed between the charges. Where, ϵ0ϵr=ϵ
ϵ is known as absolute permittivity of the medium, Hence, if F' represents the Coulomb's force in the presence of dielectric medium then,

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$F' = \frac{1}{\epsilon_r}(F)$$

$$\epsilon_r = \frac{F}{F'}$$

Above equation shows that the presence of medium other than air or vacuum reduces the electrostatic force between the electric charges by factor ϵr , which is called dielectric constant. Since the value of ϵr is always greater than one, therefore, F'<F

ELECTRIC FIELD: "The space or region around any electrical charge in which electric force is experienced is called 'electric field'".

ELECTRIC FIELD INTENSITY (Strength of electric field):

Definition: "The electrostatic force experienced by unit positive charge in electric field is called electric field intensity."

E.F.I is represented by E.

It is a vector quantity.

The direction of E.F.I is the same as the direction of electrostatic force.

Formula: If q0 is a test charge placed in electric field, then:

$$E = \frac{F}{q_0}$$

Unit: In S.I system the unit of E.F.I is NC^{-1} (Volt per meter).

Direction:

- If q is a +Ve charge, the field is directed radially away from it.
- If q is a -Ve charge, the field is directed radially towards the charge.

ELECTRIC FIELD INTENSITY DUE TO POINT CHARGE:

Consider a test charge q0 is placed at distance r in uniform electric field of source charge q, the test charge experiences Coulomb's force which is given by:

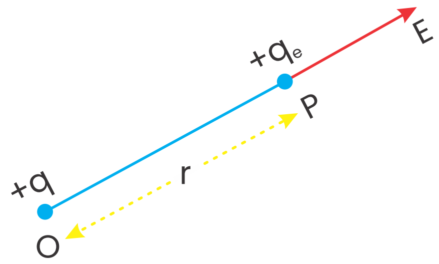
$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\hat{r}$$

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

In vector form,

$$\vec{E} = \frac{q}{4\pi\varepsilon_0 r^2}$$

$$\vec{E} \propto \frac{1}{r^2}$$



Above equations give the electric field intensity in the presence of free space or vacuumed.

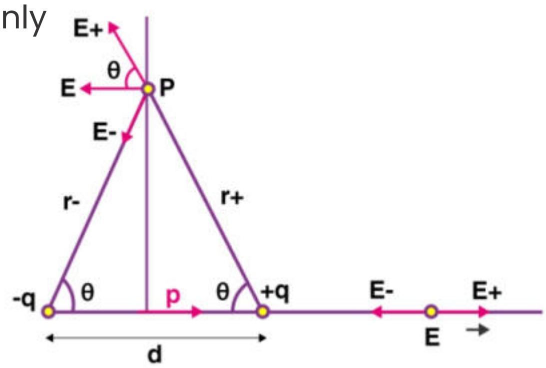
ELECTRIC DIPOLE:

"A couple of opposite charges q and -q separated by a distance d is called dipole."

ELECTRIC FIELD INTENSITY DUE TO ELECTRIC DIPOLE:

Consider two point charges +q and -q are separated by distance 'd'. Let point a test charge be placed at point 'P' which is at distance 'r' from each charge. E.F.I due to +q and -q charges move the testcharge along a horizontal line. The electric field intensities at point'P' due to dipole are E+ and E−. Each intensity is divided into its rectangular components. The sine component of each fieldcancels each other and only cosine components act along the same direction.

Total electric field intensity at point 'P' due to dipole is:



$$\vec{E} = \mathrm{E}_{+}\cos\theta + \mathrm{E}_{-}\cos\theta$$

$$E_{+} = E_{-} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

$$\mathrm{E} = 2 \times \left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\right)\cos\theta$$

From figure, In triangle ADC,

$$\cos\theta = \frac{d/2}{r} = \frac{d}{2r}$$

Putting the value of $\cos\theta$, we get.

$$\mathbf{E} = 2 \times \left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\right)\frac{d}{2r}$$

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{qd}{r^3}$$

From figure, In triangle ADC,

$$r = \sqrt{y^2 + \left({}^{d}/_{2}\right)^2}$$

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{P}{\left(y^2 + \left({}^{d}/_{2}\right)^2\right)^{3/2}}$$

where the product of the "q" and "d" is known as the electric dipole moment "P" of the dipole. The electric dipole moment is a vector quantity and it is always directed from the negative to the positive charge.

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{P}{y^3\left(1 + \left({}^{d}/_{2y}\right)^2\right)^{3/2}}$$

As $d \ll y$, therefore approximate electric field intensity of the dipole can be calculated by neglecting the term $\left({}^{d}/_{2y}\right)^2$.

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{P}{y^3}$$

As $y \longrightarrow \infty$, therefore approximate electric field intensity of the dipole can be calculated by neglecting the term $d^2/4y^2$.

$$E = \frac{KP}{\{y^3(0+1)\}^{3/2}}$$

$$E = \frac{KP}{\{y^3(1)\}^{3/2}}$$

$$E = \frac{KP}{y^3}$$

ELECTRIC FLUX:

"Electric flux is the dot product of electric intensity and vector area."

- Flux is a Latin word that means "flow".
- It is the scalar property of the vector field.
- The direction of area vector is outward perpendicular on surface.
- It is represented by φ.

Formula:

$$\Delta\varphi=\vec{E}\cdot\vec{A}$$

or

$$\Delta\varphi=E.A\cos\theta$$

Unit: In M.K.S or S.I system, the unit of electric flux is $N\cdot m^2\cdot C^{-1}$.

Cases:

- Maximum electric flux:

Flux will be maximum when vector area $\vec{A}$ is parallel to the electric field E, i.e., $\theta=0_o$.

- Minimum electric flux:

Flux will be minimum when vector area A is perpendicular to the electric field E, i.e., $\theta=90^\circ$.

- Negative electric flux:

Flux will be negative when vector area $\vec{A}$ is antiparallel to the electric field E, i.e., $\theta=180^\circ$.

ELECTRIC FLUX DUE TO A POINT CHARGE:

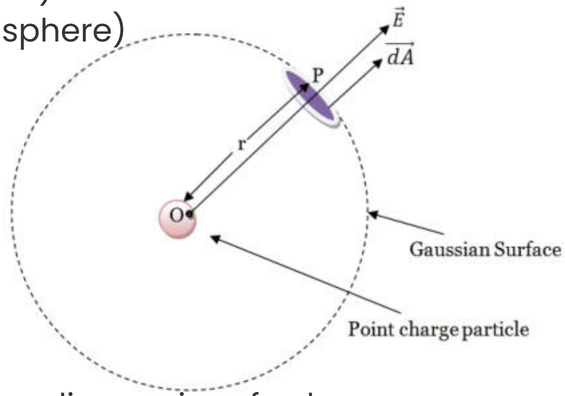
Consider a hollow sphere of radius r. Let a charge +q be placed at the center. Let the spherical surface be divided into small and equal 'n' patches, each of area ΔA, the lines of force of q spread uniformly and radially. Then flux over the surface of sphere is:

$$\phi_1=E.\Delta A_1$$
$$\phi_0=E.\Delta A_2$$
$$\phi_n=E.\Delta A_n$$

Total flux passing through the surface is:

$$\phi=\phi1+\phi2+\cdots+\phi n=E\Delta A1+E\Delta A2+\cdots+E\Delta An$$
$$\phi=E(\Delta A1+\Delta A2+\cdots+\Delta An)$$
$$\phi=E\Sigma\Delta A=E(\text{Total area of sphere})$$
$$\phi=E(4\pi r^2)$$

From the definition of electric field intensity:



$$\phi=\frac{q}{4\pi\epsilon_0 r^2}$$
$$\phi=\frac{q}{\epsilon 0}$$

Above equation shows that:

- The flux through sphere does not depend upon the radius or size of sphere.
- The flux depends upon the magnitude of charge and free space.

GAUSS'S LAW:

Introduction: The relation between electric flux and net charge enclosed by a surface was enunciated by K.F. Gauss (1755–1875), a German physicist and one of the greatest mathematicians of all time.

$$\phi = \frac{1}{\epsilon_0} \times q$$

Conclusions:

• Electric flux through a closed surface is independent of the number of charges.
• The flux through a closed sphere is independent of the way in which the charge is distributed.
• Flux is independent of the geometry of the closed surface.
• Flux depends on the medium.
• Gauss's law is applicable only to the closed surface containing several charges.

ELECTRIC POTENTIAL:

"The work done per unit charge in moving a unit positive charge from one point to another against the electric field is the electric potential."

• Electric potential is a scalar quantity.
• It is represented by 'V'.

Formula:

$$Electric\ potential = \frac{Work\ done}{Unit\ charge}$$

Unit: In M.K.S/S.I. system, the unit of electric potential is J/C=Volt(V).

Electric potential energy: *"The amount of energy required to move unit positive charge against the electric field is called electric potential energy."*

• Change of potential energy is equal to the work done of charge.
• It is represented by 'U'.

Potential difference: "The difference of electric potential between two points is called potential difference."

Mathematically:

$$\Delta U = W_{pq}$$

$$\Delta V = \frac{\Delta U}{q_0}$$

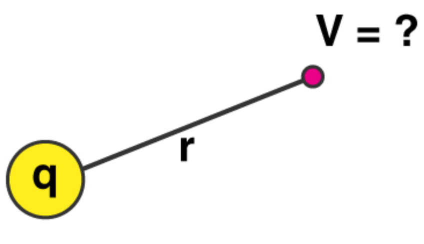
ELECTRIC POTENTIAL DUE TO POINT CHARGE:

Consider a charge 'q' is placed in an electric field having intensity.
Let charge 'q' is displaced from point A to B in the field, the force experienced by this charge in an electric field is given by:

$$F = qE$$

The work done by the charge in the field is:

$$W = F \cdot \vec{d}$$



Where $F = Kqq_0/r^2$ and $d = r$

$$W = \frac{Kqq^0}{r^2} \times r$$

$$\frac{W}{q^0} = \frac{Kq}{r}$$

W/q_0 is the electrostatic potential V,

$$V = \frac{Kq}{r}$$

Relation Between Electric Potential and Electric Field Intensity:

Consider a charge 'q' is placed in an electric field having intensity E. Let charge 'q' be displaced from point A to B in the field, the force experienced by this charge in an electric field is given by:

$$F=qE$$

The work done by the charge in the field is:

$$W=F\cdot\vec{d}$$
$$W=q_0E\cdot\Delta r$$
$$\frac{W}{q0}=E\cdot\Delta r$$

From the definition of electric potential W/q0 = V

$$\Delta V=E\cdot\Delta r$$

Above equation defines electric potential as the dot product of electric field intensity and radial displacement:

$$\Delta V=E\Delta r\cos\theta$$

The displacement is opposite to the direction of the field therefore, $\theta=180^\circ$:

$$\Delta V=E\Delta r\cos(180^\circ)$$
$$\Delta V=E\Delta r(-1)$$
$$\Delta V=-E\Delta r$$
$$E=-\frac{\Delta V}{\Delta r}$$

This equation shows that electric field intensity is the negative potential gradient.

Another unit of electric intensity:

We define another unit of electric potential as Volt/Meter or V/m = Vm^{-1}.

Equivalence of two units of electric intensity:

The S.I. unit of electric intensity is:

$$E=\frac{Volt}{\mathrm{m}}=\frac{Joule}{\mathrm{Coulomb}}=\frac{1}{\mathrm{m}}$$
$$\frac{Volt}{\mathrm{m}}=\frac{J}{\mathrm{C.m}}=\frac{N.m}{\mathrm{C.m}}$$
$$\frac{Volt}{\mathrm{m}}=\frac{N}{\mathrm{C}}$$
$$V\cdot m^{-1}=N\cdot C^{-1}$$